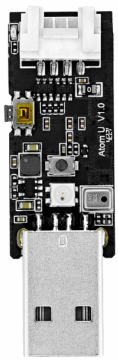
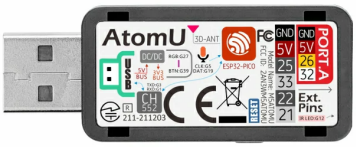
AtomU

SKU:K117













Description

AtomU is an ultra-compact and flexible IoT voice-recognition development board powered by Espressif' s **ESP32**. It is equipped with two low-power **Xtensa® 32-bit LX6** microprocessors running at up to **240 MHz**. The board integrates peripherals such as a USB Type-A interface, IR transmitter, and a programmable RGB LED, allowing true plug-and-play use for uploading, downloading, and debugging firmware.

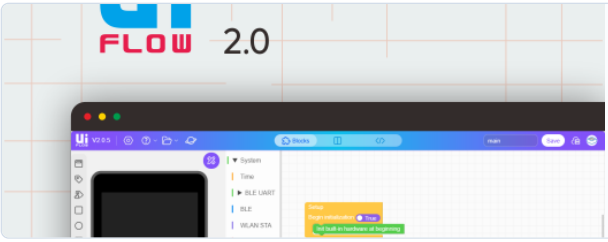
With its built-in **Wi-Fi** module and on-board digital microphone SPM1423 (PDM), AtomU enables clear audio recording, making it ideal for human–machine interaction and speech-to-text (STT) applications in various IoT scenarios.

Tutorial



UiFlow1

This tutorial shows you how to control the AtomU device via the UiFlow1 graphical programming platform.



UiFlow2

This tutorial shows you how to control the AtomU device via the UiFlow2 graphical programming platform.

Features

- ESP32-PICO-D4 (2.4 GHz Wi-Fi)
- USB-A programming/power interface
- Integrated programmable RGB LED and button
- Ultra-compact body
- Built-in IR transmitter
- Expandable pins and GROVE interface
- Development Platform
  - UiFlow1
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x AtomU

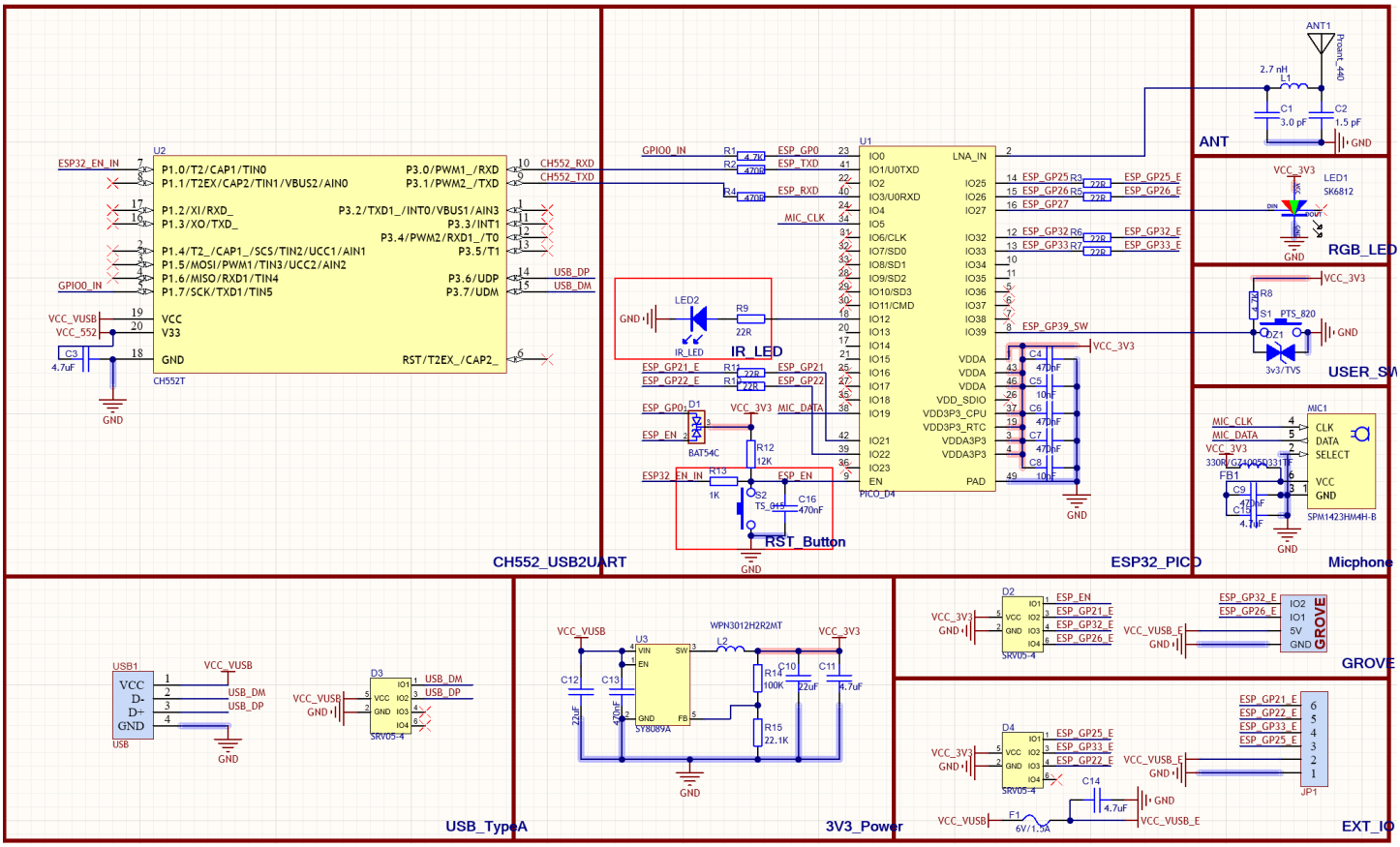
Applications

- IoT controller
- Voice recording & cloud STT recognition

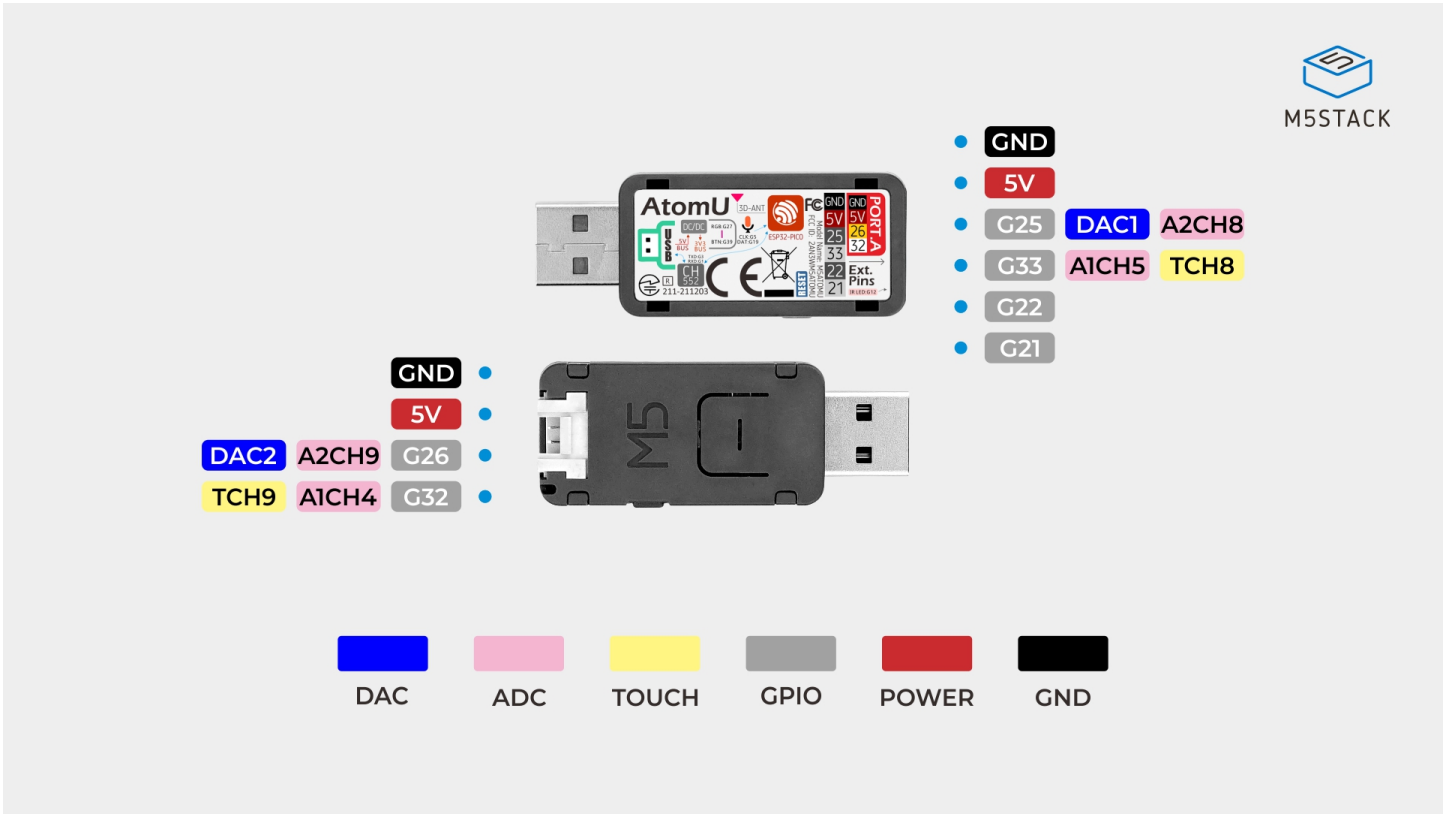
Specifications

Specification	Parameter
SoC	ESP32-PICO-D4 @ Xtensa® 32-bit LX6 dual-core processor, clock frequency up to 240MHz
DMIPS	600
SRAM	520KB
Wi-Fi	2.4 GHz Wi-Fi
Microphone	SPM1423
Microphone Sensitivity	94 dB SPL @ 1 kHz, Typ.: −22 dBFS
Microphone SNR	94 dB SPL @ 1 kHz, A-weighted, Typ.: 61.4 dB
Standby Operating Current	40.4 mA
Supported Input Frequency	100 Hz ~ 10 kHz
Supported PDM Clock	1.0 ~ 3.25 MHz
Product Size	53.0 x 20.0 x 10.3mm
Product Weight	8.6 g
Package Size	114.2 x 65.6 x 20.6mm
Gross Weight	16.6 g

Schematics



PinMap



SPM1423 – PDM

ESP32-PICO-D4	G5	G19	3.3 V	GND
SPM1423	MIC_CLK	MIC_DATA	VCC	GND

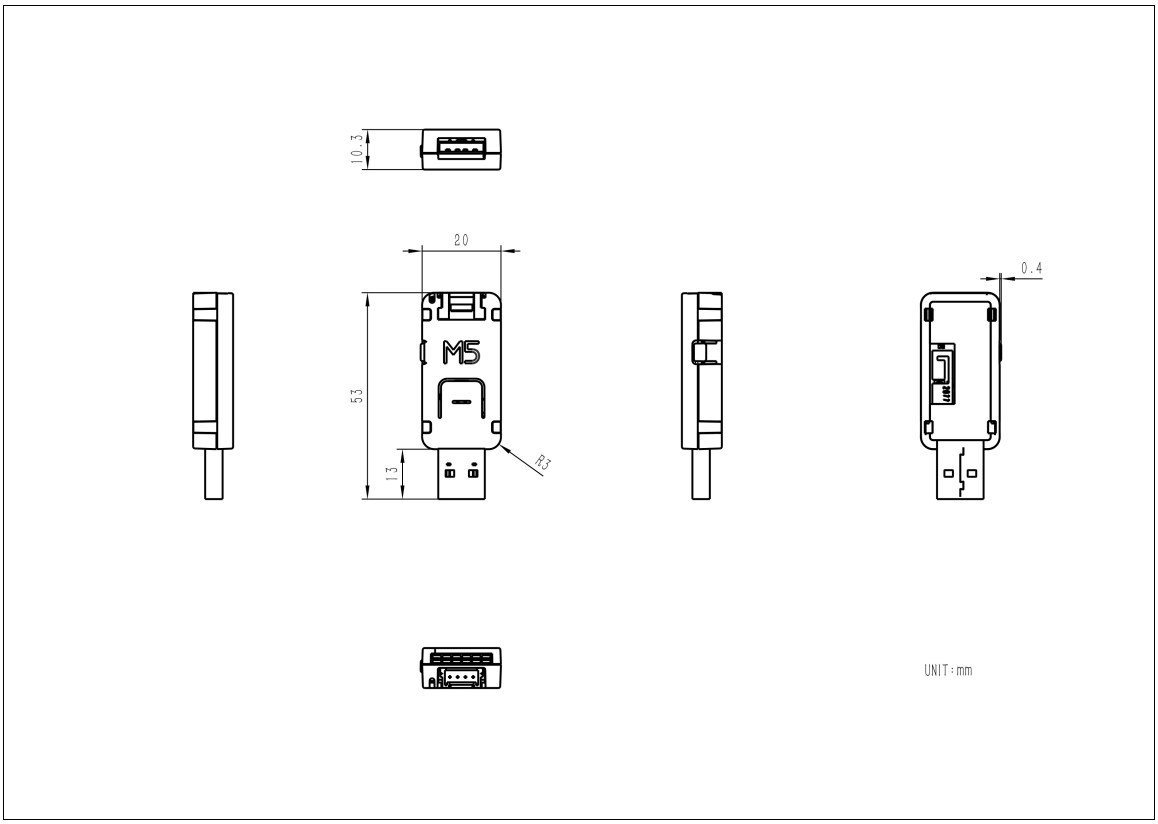
IR & SK6812 & BUTTON

ESP32-PICO-D4	G12	G27	G39
IR	TX		
SK6812		DIN	
BUTTON			SW

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G26	G32

Model Size



Datasheets

- ESP32-PICO
- SPM1423

Softwares

Arduino

- AtomU STT Example

UiFlow1

- AtomU UiFlow1 Quick Start

UiFlow2

- AtomU UiFlow2 Quick Start

PlatformIO

```
[env:m5stack-atom]
platform = espressif32@6.7.0
board = m5stack-atom
framework = arduino
upload_speed = 1500000
monitor_speed = 115200
build_flags =
    -DCORE_DEBUG_LEVEL=5
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

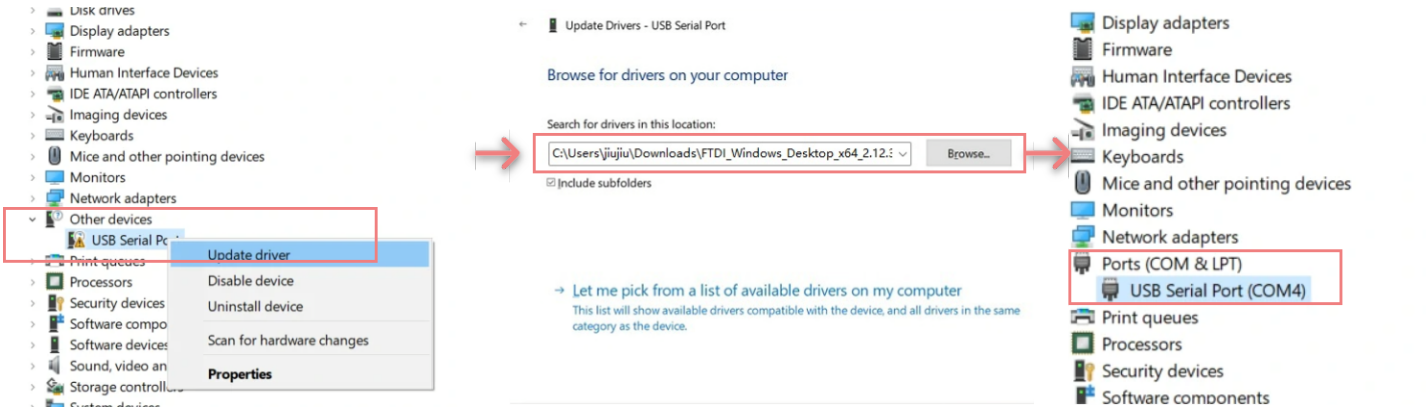
USB Driver

AtomU may not work without a driver on some systems. You can resolve this issue by manually installing the FTDI VCP driver.

1. Select the driver file that matches your operating system from the list below, download it, and extract it. You can also visit the FTDI Chip official website for additional FTDI driver versions.

Driver Name	Download Link
Windows_Desktop_x64	Download
Windows_Desktop_x86_32	Download
Windows_Desktop_ARM	Download
MacOS_Intel	Download
MacOS_ARM64	Download

2. Connect the device to your PC, open Device Manager, and install the FTDI driver (the screenshot below uses Windows 10 as an example). (Note: On some systems, you may need to install the driver twice for it to take effect. The unrecognized device is usually named `M5Stack` or `USB Serial`.)



Video

Product Comparison

To compare information on the Atom series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.